

ActInf Livestream #031.0 ~ "Non-equilibrium Thermodynamics and the Fre

Livestream | 1:36:07

[Laughter]

hello everyone welcome to the active

inference lab this is active inference

live stream number 31.0

it's october 14 2021.

welcome to the act imp lab we are a

participatory online lab that is

communicating learning and practicing

applied active inference you can find us

at the links here on this slide

this is recorded in an archived live

stream so please provide us with

feedback so that we can improve on our

work

all backgrounds and perspectives are

welcome here and we'll be following good

video etiquette for live streams

at this short link you can see some of

the past present and future

streams that we'll be doing in the lab

as part of the dot com's organizational

unit and the first tab has the regular

tuesday discussions that are group

discussions dot one and dot two

right now we're in a dot zero kind of an

ad hoc scheduled contextualizing live

stream

and then the other tabs have other kinds

of live streams that can be scheduled at

different times different topics so just

let us know if you want to get involved

or co-organize an event with us

today in active stream number 31.0

the goal is to learn and discuss

this paper non-equilibrium

thermodynamics and the free energy

principle in biology by patricia

palacios and matteo colombo

like all the dot zeros you'll get the

disclaimer that this is just an

introduction to some of the ideas of the

paper and the context it's not a review

or a judgment or a final word

in

31.0 we're going to start by looking

over some of the key aims and claims and

the abstract and the roadmap of the

paper to get the big picture

then talk about some of the keywords

maybe you're really familiar with some

of the keywords maybe it's the first

time you've thought about some of these

topics and then we'll walk through some

of the figures and formalisms and just

get set up in the coming weeks to be

discussing this paper

so we'll start with just some

introductions and warm-ups we'll say

hello and then maybe just mention

something that we're excited to

talk about or learn about or something

that we noticed while reading the paper

so i'm daniel i'm a researcher in

california and this is just an exciting

paper and topic because it's at the

intersection of chemistry and physics

and thermodynamics but also active

inference and it's a nice

starting point for a conversation about

what are the precursors and the tenants

and the principles of free energy

principle and active inference and then

what are the consequences and what does

that mean for the big picture blue
i'm blue knight i'm an independent
research consultant in new mexico uh and
i really was excited to see this paper
bring up some of the points that i've
brought up as i've been reading more
about active inference
specifically with uh respect to
ergodicity and
what that means the potential
implications for that
and also just other minor topics but
things i've mentioned in the past
it's cool sometimes to see how the
threads of our previous discussions and
things that
a participant might have brought up like
hey this isn't making sense and it's
like don't worry it's not just you these
are open
research questions and so we're seeing
that and we'll be enacting it in the
coming weeks
some of the big questions in the paper
kind of framed in
non-technical or in not the author's
words
it's really asking what fundamental
ideas from physics are utilized in the
free energy principle and how are they
utilized why are they utilized and then
what are some implications or trade-offs
associated with the use of these ideas
either the ways that they have been used
what are the pros and cons or what are
the implications and what about the ways
that they could be used
so in the spirit of

positive contributions to the literature
how are we taking ideas from physics and
other fields
bringing it together in free energy
principle and then what are the
fundamental or just spuriously
contemporary tradeoffs
let's first go to the aims and claims of
the authors
so
all read it and then blue you can give a
thought
well the paper is the non-equilibrium
thermodynamics and free energy principle
in biology on the fillsci archive at
pitt so check out the paper and read it
if you want to
and a few of their aims and claims that
we can extract out
the free energy principle has received a
lot of attention but its foundations in
statistical physics and dynamical
systems theory have not been probed yet
this lack of attention is unfortunate
because the theoretical adequacy of the
free energy principle as well as its
practical utility for the study of key
biological properties such as
homeostasis and robustness depend on the
validity of those foundations
here we begin to fill this gap
so that's kind of one of their
contextualizing claims
and their aim
and one of their
arguments
a claim about their arguments is that as
we are going to argue the foundations

and physics concepts allow models built from the free energy principle to achieve maximum generality but these assumptions also decrease the ability of these models to include enough factors to provide biologically plausible representations of the causal networks responsible for living systems dynamic equilibrium

any thoughts on the aims or we can continue

wait i can't hear you

i think that the aims are good sorry i just realized i was muted um i think that the these are good aims i think that uh one of the fundamental um

claims is that in order to make the fep more general generalizable it's not as specific to biology as it could be i think that that was like the key trade-off maybe

yes between specificity and utility and generalizability so

one mindset is that as we generalize we're going to be explaining more systems different systems explaining them better doing more and it's just all up up with generalization

another contrasting opinion is that as we generalize we out of necessity abstract or average out over particulars and so we might actually lose utility or potentially

and even disastrously

lose the specifics of the system that we're talking about so all of a sudden we have this free floating

generalization or abstraction that maybe initially was inspired by wanting to explain some sort of key biological property like homeostasis and robustness in the author's view but then we end up sort of um throwing out the kernel or the essence of what we wanted to explain because we generalized too much so we'll be exploring that okay do you want to read the abstract sure according to the free energy principle life is an inevitable and immersion property of any ergodic random dynamical system at non-equilibrium steady state that possesses a markov blanket first in 2013. formulating a principle for the life sciences in terms of concepts from statistical physics such as random dynamical system non-equilibrium steady state and ergodicity places substantial constraints on the theoretical and empirical study of biological systems thus far however the physics foundations of the free energy principle have received hardly any attention here we start to fill this gap and analyze some of the challenges raised by applications of statistical physics from modeling biological targets based on our analysis we conclude that model building grounded in the free energy principle exacerbates a trade-off

between generality and realism because
of a fundamental mismatch between its
physics assumptions and the properties
of actual biological targets

cool

and

there's a lot to say that's just the
opening

salvo from the authors

how do they make good on this abstract

we can look at the roadmap

there's not too many sections in this
paper but there's a lot of uh meat
within

there's a general introduction and then
a specific introduction to the free
energy principle which is described in
the context of self-organizing
dissipative systems

and then there's a little discussion on
the explanatory scope of the fep
and the introduction of the winged
snowflake

thought experiment or idea which we'll
return to soon

section 3 talks about biological systems
as random dynamical systems which isn't
something that fep invented it's not
something that active invented this is a
broader question about how to model
biological systems and dynamical systems

section 4 models biological active
states as random dynamical attractors
and talks about three big physics ideas
that if we take section three you know
if we want to be modeling biological
systems as dynamical systems then
we're using this physics ontology and

what other key terms are relevant in
physics for discussing random dynamical
systems and we will go through them
they are the idea of phase spaces in
biology
ergodicity
and random dynamical attractors there's
then a discussion and a short conclusion
so that's kind of the overall
roadmap that's where we're going to go
we're going to be exploring that
trade-off or the implications
of the introduction of
physics fundamentals
into the generalization of biological
systems and maybe even non-living
systems as well and just
how's it all going to come together and
how does it influence how we research
going forward
the key words
are
fep free energy principle
dynamic equilibrium
homeostasis
phase base
ergodicity and attractor
and we can ask where's active inference
not that it has to be a keyword for
everything but
as we
discuss let's keep in mind what are the
differences between fep
and active and what are the implications
of
these claims about fep for active
all right how about the first keyword
dynamic equilibrium

so interestingly like you um said that
you like how this paper brings together
physics and chemistry and biology and i
thought that chemistry was kind of
blatantly omitted

and in my heart i'm a chemist like if i
hadn't studied neuro i would have really
like gone into ocum i just thought that
like working in the lab was a little
scary like too much ether and two like
whoa when i came out um

so so really uh when i look up dynamic
equilibrium or the context that i'm
familiar with dynamic equilibrium is in
chemistry and so a dynamic
equilibrium occurs when you have a
reversible chemical reaction so it's not
that the

reaction is over right like here's an
example of water um ionizing into
hydroxide and um protons right
and so

you you it's not that when you have the
equilibrium that the reaction is still
occurring it's just occurring at equal
rates in the forward and reverse
directions

so this is dynamic equilibrium and it
only happens in a reversible chemical
reaction like so so there's no other way
that i know of that dynamic equilibrium
um happens it is an example of a system
particularly a chemical system at a
steady state and so i looked through the
paper i don't know if you want to flip
to the next slide

um i looked to the paper and i i looked
like well how are the authors using

dynamic equilibrium

and i found this um phrase which i'll
say here the core of this account is a
free energy principle

um according to which all biological
systems actively maintain a dynamic
equilibrium with their environment by
minimizing their free energy which
enables them to avoid a rapid decay into
an inert state of thermodynamic
equilibrium and so then there's some
citations first in and the ramstead 2018
paper and par and first in in 2019 and
so even i looked back through these
papers and looked for dynamic
equilibrium and it's not there
in the references that i was able to
find and so i was curious what the
authors meant

by dynamic equilibrium in a living
system i mean as far as i know um like
maybe they meant dynamic coupling
between the organism organism and the
niche

or maybe they mean the non-equilibrium
steady state of homeostasis which is not
a dynamic equilibrium it's it's a
non-equilibrium so i just found this
this phrasing kind of curious um
and i'm assuming just based on context
that they're referring to homeostasis
but homeostasis includes plenty of
irreversible reactions irreversible
chemical reactions and so that's kind of
why this was a little bit

it makes sense like if we think about
instead of the water ionizing and the h
plus and the oh minus coming back

together and the overall values on both sides are staying the same so let's think in the biological case like moment to moment our mass is staying the same let's just say or some people might maintain the same mass for 30 years but

underlying that stability is actually many many

you know synthesis of fatty acids and the breakdown of fatty acids and so it's like okay i get it that they're both going both sides at the same rate so there's a stable point

but then also it's constituting its own irreversible thermodynamic reactions sometimes and so it's a good question

do

fristen at all use the term dynamic equilibrium

and are we talking about dynamic couplings which might be at steady states or are we talking about the non-equilibrium steady state of homeostasis

okay

brings us to homeostasis

sure so homeostasis is the steady state of

internal internal physical and chemical conditions that are maintained by living systems and there's a lot of variables that are can be included in here uh mass i think actually changes at night like the the carbon that you exhale like you actually weigh less in the morning from like sleeping on my face and like

emitting carbon i heard um i don't know
if that's really true but i guess it
depends on what scale you're measuring
um but but i think that ph has to be the
same or within like a pretty defined
range um osmolarity sodium potassium and
calcium ions blood sugar
body temperature is a huge one
so and and we act we do things
throughout the day like eat and drink
and use the restroom and put on jackets
to maintain to help with our you know
homeostatic conditions um internally
well unless we're eating at night we're
losing water and some other molecules so
definitely we're lighter in the morning
and then one discussion that we've had
in a few different
streams is
when do we think of homeostasis as
reducing uncertainty from a set point
like sort of a desired temperature and
then we want to just reduce our
discrepancy versus when do we see
allostasis or almost like anticipatory
processes by which biological systems
actually
take excursions
qualified excursions from set points and
how do we think about that within the
broader history of cybernetics and it's
been said that um
cybernetics had some challenges with
dealing with post-homeostasis so it's
easy enough to see how a cybernetic
system seeks
a set point but then once you start
thinking what is the process by which

that set point is itself moved around
all of a sudden that challenged a lot of
historical cybernetics frameworks so
that's definitely what we're exploring
with active

let's take another dip into a physics or
a mathematical idea of a phase space
so

i'll read the yellow parts or you want
to go for it

i'll read the yellow parts and then
you'd give any description so in a phase
space every degree of freedom or
parameter of the system is represented
as an axis of a multi-dimensional space
so like every point on this spiral has
two variables that describe it it's like
its value on the x and value on the y
and then

the phase space trajectory represents
the set of states compatible with
starting from one particular initial
condition located in the full phase
space that represents the set of states
compatible with starting from any
initial condition

so just a quick comment or a thought on
that so even just in the maintenance of
homeostasis like where your you know
osmolarity is the same and your blood
glucose is the same or just
intracellular glucose or atp levels or
you know if you just want to look at at
just maintaining homeostasis not even
including like

the potential possible actions that i
could take in like moving in a
three-dimensional environment or

anything like that but but it it becomes
very quickly an expansive
multi-dimensional space like something
that we're not even like here we have
like a 3d ish visualization but you know
in all these dimensions it becomes very
um very quickly computationally
intractable because of the
dimensionality that of options that
we're presented with and just even to
maintain a homeostatic set point is is
intense

yes

base spaces are just kind of the
descriptive it's literally just the
space that something can happen in
and then there are trajectories through
phase space well there's points which
just are static values like three comma
four and then there are trajectories
which are like linkages amongst sets of
points

and then also

out there in phase space are attractors
so in the field of dynamical systems
an attractor is a set of states in the
phase space
towards which a system tends to evolve
for a wide variety of starting
conditions of the system
system values that close enough to the
attractor values remain close even if
slightly disturbed so a kind of mental
model there is like
a rubber sheet and someone pulls the
middle down a little bit that is now an
attractor because things are from near
or far are attracted towards it and then

if they're at the bottom of that well
they're going to resist being moved away
from it
and so that's sort of a simple ball
rolling into a bowl attractor state and
there's simple equations that govern
that type of movement but in higher
dimensional
phase spaces there are
different
geometries of attractors
and then also for biological systems
it
is a challenge to model using attractors
but that's what this paper is really
about can we use that model of an
attractor to talk about biological
systems and how they regulate
okay so yeah and i i just pulled this
image out actually i think it was
brought up in the t in the paper and i
don't remember if the original author
who outlined these four types of
attractor if that was kaufman i think i
feel like it was kaufman but it was
cited in this paper as well about the
four different kinds of attractors and
so that's what's shown here is the
strange attractor the point detractor
the cycle and the taurus um so just in
case you were wondering what those kind
of attractors looks like it's it's a
drawn to points in this kind of way
it is also on
another phase space of attractors
with its own x and y axis
okay
ergodicity

so this is a quote from the paper then
blue any thoughts would be cool
the ergotic density is an invariant
probability measure that can be
interpreted as the probability of
finding the target system m in any state
 x when observed at a random point in
time first in 2012.

the assumption of ergodicity is
important to get the fep off the ground
since it ensures that organisms can be
modeled as having invariant
characteristics over time

so just to kind of relate ergodicity to
the phase space

so the concept of ergodicity is in a
potential phase space

that all of the potential points in that
space are visited at one point or
another like that is ergonomicity and
then the probability that you can find a
uh

the system in that space or it at that
point in space corresponds to how much
time the system spends at that point in
that space if that makes sense it's kind
of a circular loop but

this is just one slide we're gonna
return to ergodicity later on but this
is um

a topic that blue and i like learning
about however if you know more about
this it would be great to have your
perspective as well as if you're also
just learning it and have questions to
bring because

there's a lot you could learn about
ergodicity and the ergotic hierarchy so

take a dive and then let us know how the water is

also getting fep off the ground i'm

thinking does fep start on the ground

does it start underground does it start

is it a snowflake with wings

has it always been that way

so

right exactly it's like the water cycle

okay so

on to the fep which is of course the

target the system of interest of this

paper and in active livestream 14 and

before then after then we've had a lot

of discussions what is the fvp how does

it relate to active inference and

it seems like we'll be working on those

questions for a long long time but a few

pieces that we've pulled out as kind of

being good

um sort of

cues

to bring up alongside free energy

principle and just see them as like

cousins or in a network together

multi-scale systems

ensembles like the collective behavior

of multiple particles or other

system subunits

the nexus of information and

thermodynamics which is really like

information

thermo like heat and dynamics dynamical

systems often thermo and dynamics aren't

really separated but we're getting at

that delineation

and then

action policy selection as variational

bayesian inference so using information
dynamics to study how policies are
selected
and then
they're sort of fed as it is today and
then there's where could we go and
develop with this framework which is
what we'll be exploring today
any general comments before we get into
the paper
uh only that you know you we haven't
thermodynamics has classically been one
word so like i'm starting to just use
info dynamics as one word just to be
like non-discriminatory you know or just
dynamics it's like it's all information
it's all thermo so yes exactly
okay here we go to the paper
this is the first paragraph of the paper
and it's very informative
life scientists use the term equilibrium
in various ways and so we're going to
look at three different definitions of
equilibrium
and this is one reason why it's
important to have ontologies and have
ways of using terms that we can if not
distinguish at least be clear about in
their use so
sometimes life scientists
use equilibrium to refer to an inner
state of death where the flow of matter
and energy through a biological system
stops and the system reaches a life less
state of thermodynamic equilibrium
that's sort of the ultimate equilibrium
then there's
more often they use it to mean

homeostasis which is the ability of keeping some variable in a system constant or within a specific range of values

going back to these old citations for where the term actually arises from and so that's sort of a living system so there's parts of it that are consuming energy hence not at the ultimate equilibrium but there's also other elements of it at least over certain time scales that are invariant some other time

the term equilibrium is associated with the concept of robustness which refers to the capacity of systems to dynamically preserve their characteristic structural and functional stability amid perturbations due to environmental change internal noise or genetic variation so it's sort of like a low bar a medium bar and a high bar of equilibrium ranging from just yeah the molecules are not doing any motion and there's a carcass on the ground two the molecules are churning and there's something invariant about the living system

going up into even the idea of robustness which is that the system is doing something to maintain its sort of equilibrium like its balance amidst noise which is maybe seen as a type of homeostasis but not exactly the same so any thoughts on why um these distinctions are matter are mattering or why start the paper this way

you know i'm curious daniel like i i
mean we've been in the life sciences for
20 some years more maybe um and i've
only used equilibrium in terms of life
sciences in the first way to like as
death um so i'm just curious and perhaps
this is the the inner chemist in me but
but are you familiar with all of these
uses of equilibrium and the life
scientists in the life sciences also
i think i've
seen the second one people might not
call it equilibrium but like a buffer
equilibrium well the carbonate and the
co₂ in your blood and it's maintaining a
buffered a ph equilibrium
so that's not death it's a living system
that's doing it but it's referring to
some chemical process that's being kept
at a dynamic equilibrium
within the scaffold that a living
biological system is providing
and um
i'd also be curious i don't think we
dove into it this kitano 2004
like
homeostasis to keep something within a
range of values it sounds a lot like
robustness so how are the how are the
second and the third definitions
different because like anti-fragility is
something different where perturbations
actually strengthen the system or cause
it to develop for example but robustness
is a little bit more like homeostasis
but
open to anyone's thoughts we'll see
now

on to
fristen at all
so
over the past 20 years is giving us some
historical context
since the new millennium
theoretical neuroscientist carl fristen
and collaborators have developed an
account of the conditions of possibility
of a certain kind of dynamic equilibrium
between a biological system and its
environment
the core of this account is a free
energy principle according to which all
biological systems actively maintain a
dynamic equilibrium with their
environment by minimizing their free
energy
through inference in action
which enables them to avoid a rapid
decay into an inert state of
thermodynamic equilibrium
so here's a few of the citations 2012 13
rams said at all 2018
par and first in 2019 so here are the
papes
if people want to dive into the
references those are always really good
places to start
but whether you take the authors at
their uh manifold dimensionally reduced
word or you go into these original
papers or you know you send a tweet to
an author or you come in a live stream
and talk to them
what are the key claims of the fep
from a rhetoric perspective what does
the fep assume and what does it claim

which papers
are key to fep
not what was important in 2005 from 2002
but today what is important so how do we
respect that different citations from
the past
might have a partial picture or use a
different formalism
and then which papers are contradictory
even if only temporary so somebody might
make a claim in a 2010 paper you can't
explain situation x and then like you
know two weeks later first in it all
write the response
and then
some people have uh they bought the
rumor but they haven't heard the news
so how do you
keep a coherent narrative of the
literature
as there are
errors contradictions disc concordances
all kinds of things that just make it
less easy than it could be to understand
what is happening with fep its
proponents and its detractors
um so just a thought on this and
something that i was thinking about as i
was reading the paper and a lot of this
is footnoted throughout the paper but
the fbp like has evolved over the last
20 years so it didn't start off like in
the state that it is now it didn't start
off like a completely elaborated um
theory
and so it would be really cool to
develop a timeline of the fep like first
concept you know 2006 or whatever and

then

at what point in time did like the

markov blanket come into play and then

when did active inference separate from

the fep and and to kind of get like a

good idea of this situation as it's gone

through it's several iterations now

great idea blue i hope that in actin

flab and other groups we could

collaborate to really make that happen

because to have the timeline of

developments

and cross-link that with the core terms

which we'll talk about later today and

then also

just who was involved and what were the

contributions it would just make it

tremendously easier instead of just

going back to square one sort of a

scholar search just when we're curious

about hey has there been something

written about free energy and x

okay so that was sort of the neutral

context or

at least not so

barbarous

but

here's where the authors uh bring in the

the gap the free energy differential in

the literature and their contribution

what actions are they going to take to

reduce the differential

the free energy principle has received a

lot of attention

no citations from who a lot relative to

what

it's not i'm not saying it's incorrect

in the paper just it would be awesome to

learn

what does it look like to have a lot of
attention what does a google trends
or a book engram viewer search results
in

are we on the up or are we going down
from some period of time so what
attention what regime of attention has
the fep received and from whom
but

the foundations of fep in statistical
physics and dynamical systems theory
have not been probed yet
that's a claim maybe some people will
think that they have so we'll look
forward to learning about that
this lack of attention is unfortunate
because two reasons one the theoretical
adequacy of the fep
and two its practical utility for the
study of biological properties
depend on the validity of the
foundations so it's like we want the
house to be beautiful and useful
and if the foundations are not strong it
will not be beautiful and or useful
so this work of the paper as well as all
the work around it is related to that
gap

so a few big questions like what does it
mean for a principal fep
and

we can of course
have some subtlety with like what's a
principle a framework a theory a
hypothesis so let's just
kind of roll with it together what does
it mean for a scientific principle to

have foundations in
statistical physics what does it mean
for it to be grounded in dynamical
systems
why does that matter why does it matter
how it's grounded or who thinks it's
grounded in what
for the fep and in general
and then always trying to connect the
generalities with the minute particulars
what does this paper actually do
and then what work is still needed
because it's easy to get caught up in
some of these big sweeping
multi-decade
many person
research trajectories and it's helpful
sometimes to just separate out what is
the paper doing and then what is the
bigger picture like the grand synthesis
that has to be done between fep and some
other field
anything here
okay
here's from a footnote
and it's um
to forestall any confusion okay great
all my confusions are about to be
forestalled i love footnotes like that
our focus is on free energy theorists
not principalists by the way theorists
assumptions that all biological systems
at any scale can be modeled within the
framework of ergodic random dynamical
systems and that homeostasis and or
robustness can be defined and studied in
terms of dynamical attractors
so i don't know if that's a assumption

of

all fep

colleagues i don't know if it's a um

rather than an assumption but rather a

consequence like there are assumptions

that people use but then actually they

have come to this claim as a consequence

not as an assumption or an ungrounded

axiom

and while our discussion should suggest

that concepts and mathematical

representations from statistical physics

and dynamical systems theory are

sometimes abused in biology

it should not suggest that those

concepts and representations have no use

so it's saying in fact you can have good

explanatory predictive

models that involve dynamical systems so

it's not saying

math and biology is hopeless

and then also this 2004 paper i thought

who do you cite for saying that you know

math

gets abused in biology so this is robert

may a famous biologist

in 2004

and this was just very interesting like

understanding the non-linear dynamics

and this was in the case of a virus and

health situation

just extracting some of the key memes

it's like we're going to have to learn

across systems we're not just going to

be able to have one default system

we need to think about multi-scale

systems

for applying

and educating in transdisciplinary contexts like having a understanding of viral dynamics is going to require population and ecological understanding as well as protein and then i venture to predict that the corresponding immunology text will indeed look different in 20 or even 10 years time okay well we're right between 10 and 20 and maybe immunologists are starting to go oh yeah you can't just talk about what's in the what's the white blood cell touching the pathogen there's a bigger picture there's a society and then what are those abuses and they're not always easy to recognize especially because few have the expertise or the confidence or the personal narrative that they're able to ask questions across disciplinary contexts so it's easy sometimes for something to be transdisciplinary and then like the biologists don't question the physics and the physics don't question the biology and all of a sudden something is not really getting questioned so it's totally a great point and this is a really important kind of fallacy essentially where a mathematical model is constructed with an

excruciating abundance of detail in some aspects while other facets of the problem are misty or vital parameters uncertain within a massive range and this happens just so many places it's unreal we made a mathematical model of this oh well you know of course we didn't consider other factor abc so it's not that everything has to be a theory of everything but we have to recognize that extremely partial mathematical models in biological systems may be less helpful than people claim them to be and it might be better if people pooled their efforts and worked towards better models any thoughts on that just uh you know i mean i think i'm definitely guilty of uh like i remember back to uh mike levin's live stream when when we were abusing physics um so guilty as charged sometimes um but but it's it is important to you know have an open dialogue and to ask questions like if there's a concept or something that you don't perhaps understand maybe like you feel like you're a beginner i mean i've definitely felt this way like maybe i'm just being dumb but you know and then there's actually a very valid question at the end of that sentence so if you don't understand something be sure to ask until you do i think yep

imagine just all the times somebody has
said oh fep does that apply to living
systems and so yeah it's a done deal
well you know it's not a done deal so
the conversation is still happening
here we get to section two introduce
introducing the fep the free energy
principle fep
says that unless a biological system
minimizes its surprise it will rapidly
die

now what do those citations say
do they talk about system death or do
they talk about system persistence
more specifically fep presupposes a view
of biological systems as essentially
persisters

different authors here
and foregrounds the conditions of
possibility for the persistence of
biological systems

so that's their top-level take on fep
and then to connect a few of the
keywords in

howie 2020 refers to a system's periodic
phase attractive dynamics so attractors
in the phase space that have periodicity
in the state space to define what it is
for a biological system to exist

howie writes biological in existence is
marked by a tendency to disperse
throughout all possible states in state
space

e.g a system ceases to exist as it
decomposes decays dissolves dissipates
or alliteratively dies

in contrast to exist is to revisit the
same states or their close neighborhoods

so this is kind of bringing together
what is life schrodinger's question ram
said it all 2018
with
dissipative systems and systems that
resist dissipation systems that have a
dynamic equilibrium what is the phase
space we're talking about
these are the big questions we're
getting at
and how does one disperse oneself
exactly
yes
and how i
think uh you heard that from somebody
earlier today
somebody was getting dispersed during a
practice talk it happens
and where's surprise come into play
not just the psychological like wow i
was surprised it was a surprise birthday
party but how do we go from attractors
and dynamical systems
to
this statistical bayesian concept of
surprise and attracting sets and
lower dimensional manifolds in
attracting sets make that make that
quote big one more time for me okay so
something i i thought was um
just unique or or maybe remarkable was
it says free energy theorists relate the
notion of entropy to the informatic
information theoretic quantity of
surprise but i think that this is done
before the free energy principle arrives
like so in information theory that's
claude shannon makes the relationship

between

information and entropy and surprise so

i don't think that that's a something

that the fep started doing

well

two themes we'll return to today and

again and again which is like

first this shouldn't be a paid versus

pape battle this isn't a battle royale

for whose hot take will reign supreme we

should be collaborating on bigger

research agendas and so clarifying the

contributions that different people have

made will help demystify and make active

inference and fep more accessible

because

which claims or which contributions were

made like 100 years ago not that that

makes it more valid but

weren't made recently but sometimes when

we only see recent citations it can be

easy to get swept up but actually the

authors of this paper do a good job of

going back to some original citations

but these are big eternal questions

here's a fun paragraph in the paper

um because it's sort of like uh i i

reformatted it slightly because it it

shows that we're in

synchronous and asynchronous discussion

with fristen at all

so here is like you know him

thinking slash saying first insane

by finding this is fristen's claim

pseudo first in claim not actually his

quote

by finding a suitable attractor in the

dynamic model life scientists would be

able to gain understandings of the
workings of dot

basically like this is the dream of
active inference that if we had
attractor phase base

all these terms coming together
then

we could just

make it work for diverse biological
systems

and then

the authors of the paper write to better
understand what this means

let's examine how such idealized models

are developed and why free energy

theorists opt to build their model

starting from concepts in physics

so this is like the dream of active
inference and physical groundings for
biological systems

and then the authors do the right thing

which is to um to steal person the
argument to investigate what its actual
strengths are

and why somebody would want to make that

argument or why this is something that

we're motivated to pursue instead of

just immediately critiquing it as like

not an important thing to want

okay all right

so

here is

in the red quote from the paper

so first in at all 2006

way long ago way way long ago asked us

to

ask us to imagine a snowflake endowed
with wings

this winged snowflake could move use its wings as solar reflectors or fans and exchange energy with the environment for a much longer period than we would expect of an inanimate piece of ice to keep going under identical circumstance unlike familiar snowflakes the winged snowflake can choose actions which are the means of by which different outcomes are brought about in different states the environment its wings allow the snowflake to bring about outcomes such as lowering its core temperature in response to an increase in air temperature so let's go to first in 2006 and this is figure one so we're imagining they're sort of on the left is like a traditional snowflake model they're sort of it doesn't have any action affordances so it's just being tossed around in the soup and it freezes when it's cold and then when the temperature gets below this critical phase boundary it melts and the system fails to persist so it's like you know checkpoint are we good with systems persisting versus dissipating this is kind of our simple physical example a piece of ice freezing in the sky and melting all right now let's imagine if the ice froze in a way where it had wings and

slightly different colored wings i
wondered what the symbolism was here
but it has these two different wings
and that allows it to engage in action
on the environment so for example if it
detects or predicts or acts as if it's
predicting that it's going to head
towards a warmer area that's like you
know lower down or something all of a
sudden it can flap a few times and get
into a colder area so here this
snowflake just falls but this one is
able to flap and stay cold
and so through action
it's able to persist longer
so
it's precluding a phase transition
from solid to liquid
through
this active
exchange with the environment so that's
like the winged snowflake model
that's kind of a nice thought experiment
from yonder to help us understand like
the steps from mere physical system
to systems that start to engage in
actions on the environment
especially those that keep persistence
of the system
anything here
okay
here's figure 3 from that same fristen
2006 paper
and this is
really an awesome figure
because it shows some concordances
between actinf even though it it's not
even really used in those terms so

that's why we need the timeline like is
active even mentioned in this paper in
2006

it might not be actually

but we can see there's this

perception action cycle

and this is related to a few other kinds
of action perception cycles

um footnote paper incoming

we can think of this as like a bowl

and then the ball is rolling to the

bottom of the bowl it's a

image we return to again and again now

the y-axis is the free energy and so

here's the minimum possible free energy

and it's like the system starts here

there's a change in the environment

that causes a mismatch

that mismatch

can be reduced through two different
types of updates of a model

one is an action and the other is a

perception so which order happens

details details details but the idea is

like at each loop there's an

environmental change

which is then renormalized or returned
to something

through perception and action which
relates to an act of inference

how we say that the two ways that free

energy can be minimized are through

updates of the generative model like

learning or parameter updating or

through action how do we reduce surprise

inference and action why reduce surprise

because if we have an optimistic

generative model we don't want to be
surprised about our model where we're
living so we act and we infer
so that we persist
any comments here
but it's just a cool like input could be
the state of chemoreceptors you know
already we're going from these sort of
physics formalisms and dynamical systems
formalisms towards just oh yeah you know
drop it it could be like chemical
biological
but
was it justified in 2006 well looking
back on it
not as much as it is justified today
and what will they say in 2033 looking
back on this so
all work in progress
okay
so this is from page six of the paper
and
won't uh
read the full quote because we've kind
of gotten at this trade-off between
utility
and accuracy and generalizability but
the big questions here are
what systems is the fep modeling or
describing where has it been used where
could it be used where shouldn't it be
used or can't it be used
how good is the fep for modeling or
describing these different systems
how good could it be and what would it
take to get it there
and then also an important question for
for those of you who are fep or active

fans and you have a challenging conversation

take a step back and ask your colleague what models do you prefer with similar or different explanatory scope just what else is out there because it really helps ground the discussion in what that person's understanding or preferences are

and a lot of times people will come up blank they'll say i'm critiquing your unified model of perception cognition action okay great critique away that's what i'm doing too what do you prefer up nothing so it helps us stay positive with the contributions that we're making to this field while also recognizing the discrepancy

and that combination of recognizing discrepancy and taking action is active inference

you know this is funny this is like a good contrast to the sims paper that we just did and i feel like at the end of the last live stream we were just talking about like thermodynamics and information theory and the relationship between those two things um but i always want to take the fvp and use it for like non-biological systems i think i like veer toward pan psychism more than most people

and like i don't like that i have a hard time with the line between mirror active inference and adaptive active inference or like the evolutionary late evolutionary cover slate whatever

uh like the you know brain is the
ultimate um source of cognition and
capable organ capable of cognition and
so like would the fvp apply to a set of
molecules in a beaker
like it doesn't have preferences did
molecules prefer to be in an equilibrium
state or how does that work out like i
want to take the fvp and use it
in systems that maybe
it's not designed for i don't know i
tend to want to go the other way
makes sense
here's that adaptive versus mirror
active inference slide from 30. good
segue perfect segway and um so here's a
question from
fep landia
what makes biological cognition special
or different
so
we're not going to go too much into it
here we talked about it in
live stream number 30 a ton
and it was just super interesting to
what extent does it matter that it can
engage in counter factuals does the
temporal depth matter
but if we're on the outside how do we
know what the temporal depth is is it
the temporal depth it acts as if
it's cognizing
where does energy dissipation come into
play
so these are big questions what is
cognition
and is there just one kind of cognition
and biological systems do a special

combination of it but it's different in
kind of
decimal point but not in the top level
like are they or are they really
different in type or is it such a great
quantitative difference that it's
different in type
big
questions here
more from the paper
active inference models are phase space
representations of biological systems
so that's like
a full stop moment this isn't what
biological systems are it's a phase
space representation of systems here
we're talking about biological systems
as
forming expectations over observable
external states and inferring policies
state action mappings that minimize the
expected free energy of those states
under a generative model in some
predefined markov decision process
by minimizing expected free energy the
modeled system would attain a
non-equilibrium steady state and so it
would maintain
in some sense a low entropy probability
distribution over its states
so what is active inference
what topics is it related to
what is it based upon
we look at that ball in the bowl
and it's easy to say well the way
that the imperative that's pushing the
system forward is that the ball wants to
be as low down as possible and it's

going to be minimizing its potential
energy that way

another story about the bowl is
the ball is going to end up or act as if
it's going to end up with the tightest
bound

through time on how far left and right
it goes now that also takes you right to
the bottom of the bowl but you didn't
have to mention that you wanted to go
down

it was just that the desire to minimize
the uncertainty

led to the bottom of the bowl in that
structure

and so

how do we think about some of these
different ways and what are the physics
metaphors

and then

an assumption from the paper an
assumption made by the active inference
models developed by free energy
theorists

is that target biological systems can be
represented as random dynamical systems

what does this assumption mean exactly

what does it mean and what are the
assumptions of active

what are the assumptions of other
frameworks

does active have the same assumptions as
other frameworks does it have fewer or
different does it have more assumptions
are there assumptions of active there
are consequences of other ideas or vice
versa

so these are just awesome questions that

the authors are raising and that their work is like a jumping off point into good

good i so so one question that i always have is what if the so active inference and the free energy principle presumably have the same assumptions but but do they have to

um and so that's always something i wonder

how has active inference developed this is a fun paragraph

active inference modeling evolves rapidly

eats shoots leaves

as maxwell ramstad dr maxwell ramsay

helpfully reminded us in conversation

up to roughly 2012 the dynamics of

internal and active states inactive

inference models were determined through

gradient descent on variational free energy

so let's call that phase one

before 2012.

we need a real timeline for this this is

just informal but this is why we need

the timeline

figure out if we're truly on the best

timeline

from around 2012 until

recently like 10 minutes ago or 2019

active inference models have been

equipped with algorithms for policy selection

which evaluate the average free energy

expected under each policy so expected

free energy not just variational free

energy

in their latest work
the active inference community
have endowed their models with a
recursive expected free energy
functional which enables the models to
represent target systems higher order
counterfactual beliefs
which are beliefs a system has about the
beliefs it would have as a consequence
of action so that's the deep parametric
active inference that's sophisticated
active inference that's what we spoke
about with a mental action paper sandved
smith at all
okay so those are three waves that are
important to distinguish in the
development of active inference models
because these modeling advances all seem
to make assumptions about the ergodicity
of biological systems
and the biological meaning of dynamical
attractors
the points we make in what follows may
help explain why hashtag all active
inference models thus far tend to trade
off realism for maximal generality
so they're respecting the developments
on the framework
and encompassing it summarizing it maybe
accurately maybe not
and then saying because they all make
some common assumptions
they're all victim to this trade-off of
realism for generality
or is it utility for generality which
one was it again realism utility
what are the trade-offs so what are the
waves of active inference

how does aerogenicity come into play
in these different waves
so
i know there's a lot on there but this
is really a important piece to
understand
that there's been recent developments
and um
yeah it's still in progress
okay
here um this is some technical details
that would be awesome to just
learn more from the authors or from
anyone else
who
sees this and gets excited by it
this is about some of the physics
formalizations of
noise and dynamics and phase spaces
and just
a term that i was curious about is like
a co-cycle
so it's kind of taking that idea of
returning to an attracting state and
maybe connecting it to some other ideas
like a co-cycle but
these are just
things to um
learn and discuss about
and then
this is sort of the classic bayesian
network markov blanket
is there a markov blanket that picks out
the mind
is the network of autopoietic processes
and biological systems identical to the
markov blanket of the system what is the
metaphysical status of markov blankets

and then the authors write these debates
in the metaphysics of the fep can be
left on the side since we're not
interested in metaphysics here
i thought we were having a discussion
that was meta
to physics
so
where is your metaphysics now
okay anything else there
action in the loop
given this setup free energy theorists
are interested in finding the active
states of the systems that are confined
to a bounded subset of states and remain
there indefinitely
so we want to have our internal states
be at a certain level like thinking
about our temperature being at a certain
value or anything else
and now we want to talk about the
control theory question which is what
active states
should the system engage in conditioned
on internal and sensory states
such that the internal states do remain
bounded within a subset of states
more
uh technical details if one assumes that
the dynamics of the target organism
is dynamics is ergotic
if one assumes that the dynamics of the
target organism is ergotic
it's grammatically incorrect if one
assumes that the dynamics of the target
organism are ergotic that's why you
can't read it that's that's why the
compiler failed then that's it random

dynamical attractors can be associated
with the ergodic density ρ
 p -distribution of x given m
which is proportional to the amount of
time each state is occupied by the
organism so here's where we return to
their definition of the ergodic density
as the invariant probability measure
um there's a few other terms that i know
some listeners know about whether
they'll help us with it's another
question
but what does it mean
to talk about the
lebesgue measure of the attractor
how is that connected to dynamical
systems and entropy and
whatever this integral is calculating in
the limit of time approaching infinity
but this is kind of this thing that we
see in space and in time
like
one over space like what's the tiniest
infinitesimal space
for all space or what's the tiniest
tiniest moment for all time
so
how does that connect to
action control theory
more more on that later but here even in
this um even in this slide it raises the
question
so here it says such active internal
states in an organism and so
maybe it's the way that the diagrams are
usually drawn for active inference but i
don't ever think of active states
as internal states like there's active

states sensory states internal states
external states and i always think about
the sensory and the active states as
the on the boundary between internal and
external

so

absolutely there um are internal
external sense and action states so
what are the active internal states
there's a few ways of thinking about it
the particular states are the sense
action the blanket states and the
internal states so maybe that's what's
being referred to

or

we can well when we're talking about
mental action i guess then the active
states are internal

yes

between the past and the future
but still eternal right

like in mental action right yes

with this prince with this definition of
active states in hand

so i believe that we're now working with
the fristen ontology of the action
being not the internal states but who
knows

first in 2012 finally formulates the
principle of least action as follows
the principle of least action

the internal states of an active system
so that wouldn't be the action states
that would be the internal states of a
system with on the interface having
action

they minimize surprisal I
such that the

variation δS of action S
with respect to its internal states
 r of t
existing in r vanishes and then classic
future Fristen look at that the 2100s
are going to be just so good
the fep would suffice to derive the
principle of least action and is defined
by Fristen in the past and the future as
follows
the free energy principle
technical technical technical if the
internal states minimize free energy
then the system conforms to the
principle of least action and is an
active system
this will be good to unpack and learn
more from the authors as well as from
those who studied these areas because
it's sort of something we all ask is
like wait least action but then they're
engaging in action couldn't they just do
less action wouldn't that be more of a
fulfillment of the least action
principle but
what is it actually it's like a
stationary principle it's an invariant
so there's a little bit more nuance than
just staying on the couch
and i haven't heard this term active
system before i've heard like active
inference system or inactive system but
this active system is new to me and
indeed Fristen was a very forward
thinking guy
uh he is a very forward-thinking guy all
the way into the 2100s
um

when expressed in these terms so in the terms of physics the fep provides us with an account of biological persistence as a generic type of stability

of random dynamical systems

so then they talk about first in 2012

and 13 some simulations

and then here's what the author's

challenge is and say let's then put into

better focus some of the difficulties

involved in this challenge and ask

is there any reason to believe that all

biological systems at any scale can be

modeled as random dynamical systems

and how does this maximal generality

bear on the degree of realism of the

models

so we had a few previous discussions ago

we had um we had realism we had utility

we had instrumentalism we had all these

different axes and how are we going to

manage that trade-off because sometimes

the most quote realistic description of

a system isn't the most useful or it's

not the most general so there's a lot of

trade-offs happening

how does model realism relate to

generalizability and how does that

relate to the utility of the model as

well as other attributes like its

simplicity

that's like the relationship between the

map and the territory if you wanted the

territory then you wouldn't need a map

yeah

and then almost like the compass

um or a piece of paper that just you

know says like you are here it's like
it's so general you can always break
down and that map is going to be
at least not telling a lie
you know this is your left or something
like that this is
but
is that useful
it generalizes really well but have we
lost this whole question of navigation
in the desire to find the simplest map
so something that that i wonder about
like um you know our biological systems
random dynamical systems like what's a
non-random dynamical system like a
dynamical system as opposed to
like a static system like statics
dynamics and so like what other system
could
a biological system b
like if it if it's not a random
dynamical system what is the alternate
of a hypothesis just like what you said
about like okay well what is the better
model for cognition if not you know
active inference then what model
is the better model right like so but
what about for a biological system if
it's not random dynamical system
what is the best model for a biological
system
yes
dynamical through time dynamics random
used in a lot of different ways but is
that
statistical is it stochastic where is
determinism fit into play
well let's talk about it so here's

section four and this is like some of the main body paragraphs and contributions of the paper it's the section modeling biological active states as random dynamical attractors free energy theorists account of biological systems persistence relies on three main modeling assumptions ergodicity the existence of markov blankets that imply a partition of states into external and internal the existence of random dynamical attractors in this section we concentrate on assumptions one and three and on the more fundamental challenge of defining phase bases for target biological systems in the life sciences so the three core ideas that they're saying are underlying fep work in biological systems are ergodicity markov blankets and random dynamical attractors are those the main three that's a really important question this paper is going to focus on the first and the third not going as much into the markov blanket discussion because there's been a huge amount of discussion on markov blankets and we've talked about them severally so they're going to focus on ergodicity and random dynamical attractors that's 4.2 and 3 but that's going to be introduced by a discussion on phase

spaces in biology so that's what the authors have set out to do now um shameless lab plug we're working on an active inference ontology in a participatory way that includes references definitions translations into multiple human languages it will be many things and it will scaffold educational content that will make this entire field more accessible and rigorous so terms like face space um we've talked about we talked about state space on september 13th so every week we're bringing together different perspectives and sharing resources and working towards common understandings in this document to really get at what are the core assumptions of fep or active what do these terms mean what would somebody need to learn if they wanted to understand more so it's really important work and we're definitely going to draw upon the author's contributions here as we work to scaffold an active inference ontology so that we don't get tied up in active state action state systems with action we can be clear about how we're talking about this field and in fact we really have to be anything on our ontology document before we continue back to the paper okay but this has been a fun project with

many contributors
so let's talk about phase spaces in
biology
we introduced this in the earlier
keyword section
so the phase space the system is a giant
space formed by the relevant degrees of
freedom of the target system
and then
that's the space of the possible
a point in the phase space
corresponds to microstate
which completely determines the system
in terms of the variables and parameters
required for analyzing its dynamics and
the dynamics are determined by the
equations of motion which describe the
evolution of points or trajectories in
phase space
so here's that phase-based kind of model
and then the authors are going to
suggest that there's
at least three facts
cold hard facts that make phase spaces in
biology hard to apply
because first biological systems have
many degrees of freedom it's not just a
pendulum
second the symmetries
which are invariant preserving
transformations
underlying the observed regularities in
biological phenomena are more unstable
than physics
okay
and third the phase space for many real
world biological targets is much less
stable than that for the kinds of target

systems studied in physics

so what do you think about that or about

phase spaces in biology

yeah i mean i think i mentioned the

multi-dimensionality in the beginning

when we were talking about the phase

spaces in biological systems

um and i think that it just comes to

that with any

um with any system right like you can

always include too many parameters like

you can have the territory or you can

have a dramatically reduced map and so

finding that that correct balance is

hard and if we're looking at um

you know say states of a system and

trying to predict the next state

um

what

are all of the factors conditionally

independent is there some you know

mingling of the variables there and is

it okay to just model a couple at a time

and what if we don't know if there are

some confounding variables in the system

too so it's it's a difficult task but

it's how do we start to get a grip on it

as hard yes

and a big question in the background

here is

does this matter for applications so yes

these are in principle critiques of

doing dynamical phase space models of

biology

but we know that dynamical systems

models of neural systems of

hormonal systems

ecology these are all really useful and

people use them every day so gene
regulatory networks don't forget those
how could we forget but does something
having these sort of fundamental
critiques

prevent it from being useful not always
this is a really nice um
points about the open-endedness of
biological phase space
so this is the work of longo montvale
and kaufman 2012

who said that in physics one can
pre-state the phase space for target
systems based upon stable stable
invariances and symmetries
so we can like say the ball is going to
be rolling around in the box those are
the boundaries and here's the
invariances

historical processes
studied in evolutionary and population
biology

like all these processes that we know
and love in biology
involves symmetry breaking which makes
the phase spaces structurally unstable
ever-changing and unpredictable
now unpredictable is a little bit of a
tough one because that's a continuum
what if you can predict one percent of
the variance in great evolutionary
transitions is it unpredictable or is it
predictable you know is it half full or
half empty

and then this is from the paper
because we cannot pre-state
the ever-changing phase base of
biological evolution we have no settled

relations by which we can write down the equations of motion of the ever new biologically relevant observables and parameters but that we cannot pre-state more we cannot pre-state the adaptive niche as a boundary condition so could not integrate the equations of motion even were we to have them all so it's like there's the ball rolling around on the billiards table but now there's mutations in the genome of the ball so that there's like totally new mechanisms arising in a way that might have a historicity oh yeah and the table is mutating and it's made up of other organisms so it's easy to see how we get lost in the complexity yet adaptive action exists and useful models exist so how do we reconcile the fundamental open-endedness and creativity and novelty of evolution with our desire to explain some portion of the variance so we can act effectively well and this relates like to a question that i had brought up to you um yesterday daniel i think about like what about discretizing space and so i know that like some of the physics principles rely on infinite time like the ergodicity specifically relies on the possibility of infinite time and this is all stuff that's like kind of beyond me but but what if we have a temporarily restricted moment right like what if we're just interested about now and five years from now or what if we're just

interested in now and five years ago

right and and so

in each moment is this symmetry breaking

happening or at what scale and so is

there perhaps a chunk of time a small

discrete chunk of time from you know a

microsecond to 100 years

where

these types of um inferences can be used

and where the equations are applicable

and where we can use the model with

under the you know specification that

this model is not good for the next two

million years but it will last you 20.

right like so it's a model of the phase

space of blood glucose versus ph

in a person

as measured by the devices we have it's

not of blood sugar over a million years

because that doesn't even make sense

given the system of interest so how do

we reconcile pragmatics and utility with

these really important philosophical

questions so that was four one face

spaces

word two

ergodicity

one of the assumptions of the fep is

that all biological systems are ergodic

so

paging

ramstead at all at all

this is something we really want to

unpack and understand like

how how does it

mean

ergodicity or more precisely metric

transitivity in random dynamical systems

like what is the math and the physics
and is that an assumption of the fep
what does this mean to assume that
biological systems quote are ergodic
which is different than saying

biological systems for some spatial
temporal scales can be modeled using
statistics that involve

ergodic assumptions

so there's a lot of nuance there like
whether we're making realist claims
about how biological systems are
or whether we're using statistics in an
instrumental fashion to make useful
um

models of organisms but the big
questions are like what kinds of systems
are ergodic and how

what are the implications of ergodic
assumption is it ever not an assumption
but it's actually like a consequence of
some empirical grounding

there's the ergodic hierarchy so
ergodicity it's not just like yes or no
there's a total continuum with a lot of
complex math

and then just

for everyone

how or where does this matter

for modeling under the fep or active
inference in real world situations

and they did call out in the paper um a
reference to ramstead and the local
ergodicity which is you know for a
biological organism i'm like something
like the you know boundaries in which
the organism can exist

um anyway i think uh maxwell come come

explain it to us please but um
the authors did mention that so
here's a quote from the paper
birkhoff 1931
demonstrated that if a dynamical system
is ergodic then the infinite time limit
exists and coincides with the phase
average for almost all initial
conditions little x like the realized
conditions within the bigger x like the
space of the possible
this is called the ergodic theorem
and that is going to get linked to
everything we were just discussing
surprisal minimization as ergodicity
maintenance
and then here the proof of the principle
of least action
is straightforward
okay
and rests on noting
that action
and the entropy of the ergodic density
over external states are related via the
ergodic theorem first in the past and the
future
so
pnas
birkhoff 1931
pretty interesting paper and just
it's amazing to think like how in 100
years of mathematics after this but this
is kind of that initial formalization of
what if the space average and the time
average were going to be
related to each other
so it's a cool topic we've brought up
ergodicity before and it would be really

helpful for people who know some of the technical details to help work with it

this is one um analogous to how they brought up like here's the facts that challenge phase spaces in biology so here's some facts that challenge the application of ergodic theorems to real systems

and there's good reasons the authors say to think that the theorem cannot be realistically applied in the domain of biology so

first off it's not enough to show that

the infinite limit exists it also has to be shown that the convergence rate is plausible we're going to talk about that in a second

second point comparing biological to non-biological systems

it's harder to estimate the amount of time

that they need to kind of asymptote because of the non-uniformities

another reason and this is going to be one that i know people will have things to say

why ergodicity cannot plausibly be assumed in biology

is that the ergodic theorem requires the dynamics of the system to be ergodic which means that eventually almost every point will visit every measurable region in x

we've talked about that before like but biological systems never revisit the exact same state if you think about

every molecule constituting the phase space but if the phase base is just temperature versus blood sugar yes you pretty much will revisit some points but will you revisit every point are you going to reach a value of negative 5 000 sugar and 900 degrees celsius no

we're talking about bounded states and that's kind of a nuance and then well and also it's also the arrow of time is is moving forward right it points one direction and so like you know at age 40 my skin will never have the collagen it had at age four like that's never going to happen again right so i mean and there are some things that just decrease as you age

yes

so there are yeah

let's look at this palacios 2018 paper so this is i guess an earlier paper by the first author so it kind of gives a little context

had we but world enough and time but we don't

exclamation point

colon

justifying the thermodynamic and infinite hyphen time limits in statistical mechanics

great title well punctuated uh a bevy of punctuations and this is just a very interesting paper that we skimmed over and i'm sure there's a ton to say it'd be great to

speaking with the author about it but they argue that this has there's an important consequence for the philosophical literature on infinite limits and that's basically it's not just about like where you are that's where you are in phase space it's about where you're going that's the asymptote in infinite time and it also matters how fast you're going to get there so it's like kind of the asymptote for a pane of glass is like in a puddle on the ground because it's a liquid or it's a glass but that might take so long that over some time scale it can be modeled as if it's not changing for example so that's kind of a cool notion that it's not just about where a system is or where it's being attracted towards but actually the dynamics of the approach matter a lot and that's certainly true in biological systems um the assumption of ergodicity to give a definition of equilibrium states is a controversial assumption even in the domain of physics especially due to existence of physical systems in equilibrium that have proven to be non-ergodic so i didn't find this palacios 21 citation in the biblio this is the reason why physicists and philosophers have offered alternative

approaches to equilibrium that do not
rely on ergodicity
even philosophers that defend a
quasi-ergodic approach to equilibrium
recognize that this definition of
equilibrium may plausibly apply only to
a restricted class of systems such as
gases
so
sounds
bad
but then we go okay
the authors say it doesn't mean of
course that ergodicity or a weaker
notion of local ergodicity may sometimes
be a reasonable assumption to make
so in other words it might be useful
for specifically defined systems anyway
however the fact that one cannot
extrapolate the assumption of ergodicity
to the behavior of all relevant systems
in biology cast doubt on the project of
formulating a principle that depends on
this assumption
and is both maximally general and
biologically realistic
so ideas that we've heard a little bit
about but how are we going to finesse
realism
and
generality
and grounding in physics when biological
systems might be grounded and things
that
are not fulfilled
so that's 4.2
all right
4-3 random dynamical attractors

what makes the application of fep to
account for real-world behavior
biological systems
is even more challenging beyond 4.2 and
4.1

because one also has to show the
existence of an attractor
and also justify
why certain attractors but not others
plausibly denote homeostatic states in
the target

there might be multiple attractors
and then there's this question of
chaotic systems do you want to add
anything on these

basically it's non-trivial to say that
there's an attractor and it's the right
one and it results in adaptive behavior
there's a lot that goes into it

so

but

we can learn more we'll hear more
questions later and then just to go
quickly to five and six

so five discussion

in the previous sections we've analyzed
three challenges involved in the
justification of the free energy
principle

so one was pre-stated phase spaces
for most biological systems

two

was the lack of the warrant
for making ergodicity assumptions in
biology

and three is the challenge of
identifying homeostasis
with attractors in a phase space even if

one and two were accomplished
based on our analysis
one overall conclusion is that
because of a fundamental mismatch
between its physics assumptions and the
properties of biological targets
model building grounded in the fep
achieves maximal generality
yay
for minimal biological plausibility
bat
so
how is that going to work
as we raised earlier
how
does generality versus plausibility get
traded off for real systems
in the fvp
in active and in what other frameworks
let's always have another one if people
want to or can think of one so that
we're not just asking whether it's like
as good as it could be because here's a
hint it's not as good as it could be
we're always going to see ways to
improve the theory
and the literature if only by organizing
other work and making it more accessible
so
what is its relative biological
plausibility what is its relative
generalizability and what are we
relativizing too
because as they say
like for other idealized scientific
representations even if the fep involves
simplifying distortions it does not
follow

that it must be minimally realistic and have limited explanatory or predictive power

so they're saying like it's kind of like enemy my enemy is my friend it's like saying just because you idealize and simplify

that it doesn't mean that you're gonna get explanation or prediction

so

that's their discussion

they

fall back to the three key points

that they think challenge

the usage of physics ideas in biology

that are postulated to be underlying the

fep phase spaces ergodicity and random attractors

and they're concluding that there's a mismatch between the physics assumptions and the properties of biological targets here in the implications and predictions they write

if the fep or active inference models it grounds

make simplifying distortions

distorting assumptions about the phase space of a target system the ergodicity of biological systems and the existence of an attracting set corresponding to homeostatic states for these systems

then these idealizations should earn

their keep so you know

you've been a bad theory but

it's not all over you could be a good

theory you could earn your keep maybe in the future

and then they um write

um

these idealizations should allow life scientists to attractively draw some explanation and prediction about relevant biological observables in real world systems and assess those predictions against measurements relevant to understanding some aspect of homeostatic processes in actual biological systems
decosta at all 2020.

our contention is that the idealizations made by free energy theorists do not play these practical and epistemic roles and so this is the de costa at all 2020
active inference on discrete state spaces a synthesis
and um

here are from table one here's all the applications

so it's interesting because like this pragmatic and epistemic role that's kind of the ontology of active inference that's how we talk about policy selection

and then it's like

okay looking at this list i'm seeing a lot of pragmatic and epistemic value but

what is the contention

that the idealizations themselves do not play pragmatic or epistemically valuable roles

i think it'd be good to get a little clarification because table one has a lot of and since then even there's a lot of interesting areas what do you think about that blue

yeah it's interesting like the word choice of the author here like the idealizations made by the free energy theorists theorists so what are these idealizations like the reliance on ergodicity and random dynamical systems theory like are these idealizations or is it actually using the fep and active inference for practical applications that don't play the practical and epistemic roles so i just i'm curious as to that word choice and steven in the chat wrote the use of approximation science around ergodicity could give the potential to have sufficient revisiting of states to make meaningfulness plausible how can we measure when sufficient approximation to ergodicity exists in order to make variational free energy inferences so good question we can ask and as we kind of explored revisitation is one component of ergodicity but also there's this element of re of revisiting every point and of exploring the whole phase space so as we kind of addressed what is the reality of ergodicity and applying those models to biological systems and then yeah can we use approximations and approximation science to make it work even if in the infinite perfect case it doesn't exactly seem like it would like the t-test assumes that there's an equal variance between the two groups unless you have a t-test within equal variances

yet
no two groups in reality are going to
have the exact same variance yet the
t-test works
so how can we take that same approach
and think about the systems that we're
working with here
here's their whole conclusion
in a churning mail storm
they're arguing that fep theorists have
pursued maximal generalization
by relying on physics concepts
however that means that they've
sacrificed biological realism
and the danger of sacrificing biological
realism is that you risk
minimizing explanatory and predictive
power
of those accounts
for biological systems which are
homeostatic and
manifestingly far from equilibrium in
their persistence
and their last paragraph
the fep can perhaps be better understood
as a maximally general definition of any
system that persists
but this definition does not seem to
provide us with any new insight into
biological systems
to the extent free energy theorists
treat all biological systems at any
scale as pre-specific pre-specified
generic objects with fixed
currently unknown equations of motion
their account risks missing all features
that make biological systems interesting
kinds of thermodynamic systems so it's

like you know a city where there's all kinds of intersections and then you have a grid map and so we distort the system in a way where all of a sudden we thought we were generalizing but we ended up losing touch with the system and therefore explanatory and predictive power

any thoughts on the conclusion

um no i mean i think that that

you know they have made their point

pretty clearly so um

i think the fep also has a lot of room

to grow being that it's

you know relatively not not quite 20 yet

so i think that um you know hopefully

that

some

pragmatic and epistemic value will be

put forth by

um the generalizations made by the fep

in some future model perhaps

that's almost like a deep time

sociological value

even a provocative theory which has

limited epistemic or pragmatic value not

saying the fpp is could play a role in a

broader social system

that still gets the job done so that's

kind of a cool notion

what questions for 31.1

did you put up

so i think i raised them all already so

like what is the dynamic equilibrium of

living systems as the authors state here

and as i stated in the beginning

keywords

and it says we should pay closer

attention to the fundamental differences between physics and biology like what are the differences between physics and biology like in the paper they made they went to a great length to explain like why biological systems weren't ergotic but then also like they made the point that

physics they have they have a really hard time like proving ergonomy exists at all um in any physical system so so what are what are these fundamental differences between physics and biology that that are so critical like that it seems that you know the fep kind of misses the point

um and then they refer often in the paper to active states as as internal states and i i just wonder

what like i mean i've really thought about active states as like the interface between internal and external and part of the blanket and so i just wonder why

the active states are included in internal states and their you know description of the model so so just some some technical things that i have questions about and um you know hopefully the authors will be here to answer them cool

stephen wrote

we may never know the whole phase space as the phase space may be variable yes that's what we talked about like the phase base of temperature does that include

impossible temperatures that the system

cannot persist at and then the fact that
the phase space can change is the
open-endedness
of the um the longo i think 2012
open-endedness of the phase space in
biological systems so for sure that's
why we can read the papes and then have
specific citations and ontologies in
mind so we're not kind of cross angling
because those are great points and they
were the points made in the paper
and then stephen asks is biological
realism based upon folk biological
assumptions
so i think this sort of like folk
hyphenated everything has come into the
forefront with the discussion with matt
sims
so yeah
what is
biologically real
you know what models are biologically
realistic isn't that just like the
question of what is real
so yeah what is even real at all exactly
yeah realism sounds good until you
so you ask too many questions
so
what a fun discussion
so we ask all these questions like
what does a good understanding enable
what are the unique predictions and
implications of this paper
what are the next steps for fep and
active
what are the goals of this research
and then what are we still curious about
it was a

fun discussion so thanks a ton blue for
doing it and for working a ton on the
slides

we'll look forward to

31.1 and beyond

awesome thanks bye

bye